

Framework for preventing and evaluating potential adverse effects of using the Informed Health Choices secondary school resources (Version 1)

Background

Researchers and others often overlook potential adverse effects of educational and public health interventions (Lorenc and Oliver, 2014; Bonell *et al.*, 2015; Zhao, 2017, 2018). We are developing a framework to help us prevent and evaluate potential adverse effects of using the Informed Health Choices (IHC) secondary school resources, as well as theoretical mechanisms for each outcome in the framework.

Our starting points have been discussions within the IHC team, as well as relevant literature, in particular a framework of potential adverse effects of the IHC primary school resources (Nsangi *et al.*, 2019); a generic framework of potential adverse effects of public health interventions (Lorenc and Oliver, 2014); and suggested approaches to theorising potential adverse effects (creating a “dark logic” model) for evaluations of public health interventions (Bonell *et al.*, 2015).

Definition

An “adverse effect” is an increase in undesirable outcomes or decrease in desirable outcomes attributed to an intervention (www.getitglossary.org) (Moberg *et al.*, 2018).

Objectives

Primary: Inform the development and evaluation of the IHC secondary school resources, in Kenya, Rwanda, and Uganda

Secondary: Provide a starting point for a generic framework for preventing and evaluating potential adverse effects of critical thinking interventions

Criteria for a sensible framework:

The following criteria are based on Feinstein’s “sensitivity” criteria for health indexes (Feinstein, 1987):

- The objectives of the framework are clear and meaningful.
- The framework includes all potential adverse effects that are relevant, logical, conceivable, and important.
- The framework excludes any potential effects that are not adverse, or are irrelevant, illogical, inconceivable, or unimportant.
- The structure, organisation and presentation of the framework are clear and logical.
- The framework is as simple as possible without being oversimple.
- The theoretical mechanisms are clear, logical, and consistent with any relevant evidence.

Framework (with links to theoretical mechanisms)

Category of effect ¹		Adverse outcome ²		Converse benefit
Decision making harms	Behaviours	Mis-transfer of learning to daily life	Unintentional misapplication of concepts	Appropriate transfer of learning to daily life
			Unnecessary application of concepts (“overtransfer”)	
			Intentional misapplication of concepts	
	Beliefs	Misinterpretation of examples		Correct interpretation of examples
		Overconfidence in personal ability		Improved, but appropriate confidence
		Distrust of health professionals and researchers		General healthy scepticism
		Trust in charlatans		
Psychological harms		Nihilism or cynicism		Enthusiasm
		Cognitive dissonance		Openness
		Work-related stress		Energy and efficiency
Equity harms		Inequitable learning		Equitable learning
Group and social harms		Conflict	Student-teacher	Engaging discussion
			Student/teacher-family/community	
			Student/teacher-health professional	
Opportunity cost harms		Waste of time and resources		Worthwhile use of time and resources

¹The directly affected populations are students and teachers (participants/users), while indirectly affected populations are their families and communities, and health systems.

²The potential adverse effect is an increase in the given outcome.

Theoretical mechanisms

Decision making harms

This category replaces the category “Direct harms” in the aforementioned generic framework of potential adverse effects of public health interventions (Lorenc and Oliver, 2014). In other words, we assume that using the IHC secondary school resources will not cause or contribute substantially to direct harms to physical health.

However, it is conceivable that the resources will cause behaviours or beliefs that contribute to poor health choices, which in turn cause unnecessary harm to physical health. These behaviours and beliefs relate to how students and teachers make choices about interventions, and what they do with information about the effects of interventions.

Mis-transfer of learning to daily life

The IHC secondary school resources will focus on a selection of IHC Key Concepts and specific hypothetical scenarios. We intend for students to use resources in the classroom, for teachers to guide students, and for students to mostly work together in groups.

However, if students are only able to apply the particular concepts to particular scenarios, in a particular context, the value of what they learn is limited.

Our goal is for students to appropriately “transfer” what they learn (Barnett and Ceci, 2002) from the lessons to other contexts—especially to their daily lives. It is conceivable that at least some students will also be able to transfer what they learn to other domains, for example the environment. We are developing a separate framework to help us achieve and evaluate desirable transfer of learning.

However, it is possible for students to inappropriately transfer what they learn or what they mis-learn. Here we call this “mis-transfer”. We specify three types of mis-transfer and discuss them in turn:

1. unintentional misapplication of a concept,
2. unnecessary application of a concept (“overtransfer”), and
3. intentional misapplication of a concept.

There are different types of potential problems with the resources that might give students or teachers an incorrect understanding of a concept, leading to unintentional misapplication. These include errors or inaccuracies in the content of the resources; lack of clarity or nuance in the content; an overwhelming amount of content; and flaws in the design or technical solution.

If students learn to misapply a concept in the classroom, they might transfer this misapplication to their daily lives. For example, if the resources inadequately explain what

constitutes a “dramatic” effect, students might assume an intervention has a dramatic helpful effect, when in fact it might not have an important helpful effect or even have a paradoxical harmful effect.

Taking an iterative, human-centred design approach, we expect to avoid problems with the resources like those mentioned. However, it is conceivable that there nonetheless will be such problems—especially due to limited time and resources, both on our end and the end of the users.

Then there is “overtransfer” (Barnett and Ceci, 2002): transferring a correct application of a concept from the classroom to daily life, but applying it unnecessarily. We all make many health choices every day, and it is impractical and unfeasible to apply every IHC Key Concept to every choice—for example questioning the evidence about the health effects of every single food that we eat. In other words, overtransfer means overcomplicating choices.

Finally, it is possible for users to intentionally misapply concepts that they have mastered, to mislead others. For example, is possible to misapply the concept about randomisation to claims that smoking cigarettes causes lung cancer based on non-randomised trials, to disingenuously argue that smoking cigarettes is or might be safe.

Note that mis-transfer can also happen across domains (fields). For example, similar to the example above about smoking cigarettes, it is possible to intentionally misapply concepts to disingenuously argue for inaction on climate change (Aronson *et al.*, 2019).

Misinterpretation of examples

As opposed to many other public health interventions, such as campaigns to stop or prevent people from smoking cigarettes, the IHC secondary school resources are aimed at helping people learn how to make informed choices, not encouraging people to make particular choices. It is conceivable that students and teachers nonetheless will interpret the examples in the resources as health advice.

This is a common problem that we have found in user-testing prototypes of both the primary and secondary school resources. It could contribute to poor health choices if users misunderstand how an example applies to their daily lives (e.g. mistaking an intervention as relevant to them), or misunderstand the example entirely (e.g. mistaking an example of a harmful effect as a helpful effect, or vice versa).

Overconfidence in personal ability

The IHC primary school resources had a large effect on children’s and teachers’ ability to apply the IHC Key Concepts explicitly addressed in those resources (Nsangi *et al.*, 2020).

However, the primary school resources were not 100% effective, and it is almost certain that the IHC secondary school resources will not be 100% effective either.

Furthermore, like the primary school resources, the secondary school resources will address a minority of the concepts in the IHC Key Concepts framework, which are all essential—not to mention important skills for making informed choices that are outside the scope of the framework, such as being able to search for relevant and reliable health information.

After using the resources, students and teachers should have an improved ability to make informed choices, and a correspondingly improved confidence in their ability. They should have a general awareness of limits to their ability (that there is always more to learn), as well as any specific limits mentioned in the resources (e.g., that they have not learned about how to search for information). And they should be aware of external challenges to making informed choices, such as peer pressure.

It is possible that students and teachers will be able to “transfer” learning about the IHC Key Concepts explicitly addressed in the resources to other concepts. For example, assume the resources address the importance of randomisation and have an effect on a student’s ability to apply this concept. It is possible that she will then be able to apply the closely related concept of correlation not implying causation, even if we do not explicitly explain the latter concept in the resources.

However, in a limited number of lessons, it is infeasible and inexpedient to even mention all the concepts and skills that are necessary to make informed choices in any given situation. And there is important uncertainty about how to achieve transfer of learning (Barnett and Ceci, 2002).

All in all, it is likely that students and teachers will be, to some extent, unaware of important IHC Key Concepts and skills that they have not mastered. It is conceivable that this would lead to overconfidence in the important, but limited skills that they have gained. This might prevent them from seeking necessary information or help that they might otherwise have sought when faced with a claim or choice.

The idea of the resources leading to overconfidence comes from discussions within the IHC Team. One could argue that it conflicts with the logic and evidence supporting the Dunning-Kruger effect (Dunning, 2011). This theory suggests that mastering IHC Key Concepts could temper students’ and teachers’ confidence, or even contribute to nihilism or cynicism (see [Nihilism or cynicism](#)).

Distrust of health professionals and researchers

The resources will include examples of unreliable claims that have been made by health professionals and researchers. Indeed, some health professionals have not mastered all of the IHC Key Concepts covered in the secondary school resources, not to mention other important IHC Key Concepts (Hoffmann and Del Mar, 2017; Austvoll-Dahlgren *et al.*, 2020). We can address this problem explicitly in the resources. Students and teachers might recognise it regardless.

Furthermore, it is challenging to use examples of unreliable claims about familiar “traditional” treatments. Students and teachers might have deep-rooted beliefs in such claims, and an excessive scepticism towards “Western” treatments or the pharmaceutical industry. They might become defensive if we say that a claim about a particular traditional treatment is unreliable, or that the treatment is wasteful or harmful, and their defensiveness might prevent learning.

This challenge is exacerbated by a lack of research evaluating the effects of local or regional traditional treatments. To show students and teachers that the resources and concepts are important to them in their daily lives, we want to use real examples about familiar interventions. When selecting real examples of unreliable research, there is a larger pool of examples about “modern” treatments, compared to traditional treatments. Hence, we are more likely to use examples of unreliable research about modern treatments, which might cause or contribute to the resources disproportionately focusing on wasteful and harmful modern treatments.

After using the resources, students and teachers should have a general healthy scepticism towards all claims, regardless of the source of the claim or the category of health action. However, given the issues above, related to examples in the resources, it is conceivable that the resources could cause or contribute to excessive scepticism towards health professionals and researchers (i.e. distrust of health professionals and researchers), resulting in poor health choices, as well as spread of misinformation. This distrust would be a form of cynicism (see [Nihilism or cynicism](#))

We identified distrust of health professionals as a potential effect of the IHC primary school resources, in the process evaluation for the trial of the those resources (Nsangi *et al.*, 2019). However, we found no cases of it in either the process evaluation or 1-year follow-up of the trial (Nsangi *et al.*, 2020).

Trust in charlatans

The same mechanisms that might cause or contribute to distrust of health professionals and researchers might also cause or contribute to a lack of scepticism towards charlatans (i.e.

trust in charlatans), such as herbalists, likewise resulting in poor health choices and spread of misinformation.

Psychological harms

Nihilism or cynicism

After using the resources, students and teachers should feel good about their improved ability to assess claims and make informed choices, and their potential to learn more. Moreover, they should feel good about the past and potential benefits of reliable research. However, this involves accepting realities that can easily be disheartening:

- Many common health interventions are harmful or wasteful.
- Many health claims are unreliable.
- Many people have not learned to think critically about health, including many parents and health professionals.
- A lot of health research is unreliable.
- There is important uncertainty about the effects of many health interventions.
- Citizens have limited influence over decisions about health policies and systems.
- Students will have more to learn within critical thinking about health, but probably few opportunities to learn more in school, if any.

Having improved their critical thinking skills, students and teachers might also have a heightened awareness of what they remain unable to do—i.e., there might be a Dunning-Kruger effect (Dunning, 2011). Altogether, it is conceivable that using the resources will cause students and teachers to feel nihilistic or cynical, including distrustful of health professionals and researchers.

We identified nihilism and cynicism, as well as shortened enjoyment of the innocence of childhood, as potential effects of the primary school resources, in the process evaluation for the trial of those resources (Nsangi *et al.*, 2019). However, we found no cases of it the process evaluation, nor the 1-year follow-up of the trial (Nsangi *et al.*, 2020). Here, we consider shortened enjoyment of the innocence of childhood as a consequence of cynicism or nihilism.

Cognitive dissonance

After using the resources, students and teachers should be open to questioning the reliability of any claim or belief. Further, they should accept uncertainty about the effects of health interventions. However, what they learn could conceivably lead to an uncomfortable experience of “cognitive dissonance” (Harmon-Jones and Mills, 2019), if they are able to apply their skills to some claims, but struggle to apply them to others, for example because of strong personal beliefs.

Work-related stress

Students and teachers should find using the IHC secondary school resources purposeful and enjoyable. The resources should be designed so that the lessons are easy to prepare and teach, and to complete within the allotted time. Moreover, students and teachers might be able to transfer helpful learning and teaching strategies used in the IHC lessons (e.g. concept mapping) to other lessons and across subjects.

However, students and teachers have busy schedules. Importantly, there is pressure to focus on exams that determine academic and career opportunities for students, performance appraisals of teachers, and reputations of schools—exams that do not require the ability to apply IHC Key Concepts or even think critically (Mugisha *et al.*, 2021). Moreover, it is possible that the teaching and learning strategies, as well as some design features and technical functionality within the resources, will be foreign to students and teachers, and require substantial time to master, in addition to the content (the IHC Key Concepts).

Hence, it is possible that using the resources and teaching the lessons will be experienced as a large amount of additional, inessential and difficult work, causing stress. Teachers might also be stressed by the prospect of increased questioning from students.

In the process evaluation for the trial of the IHC primary school resources—although teachers said they found using the primary school resources enjoyable—some reported experiencing stress from the IHC lessons coming on top of the work that they already had to do, and from the content being new (Nsangi *et al.*, 2019).

Equity harms

Inequitable learning

Using the IHC secondary school resources should help students master IHC Key Concepts regardless of their initial academic advantages or disadvantages. However, it is conceivable that the resources will be most helpful to students who are at an advantage to begin with, thereby increasing inequity.

Group and social harms

Conflict

Using the IHC secondary school resources should cause students and teachers to become more engaged in discussions about claims and choices. This includes discussions between students and teachers; between students/teachers and family members or other people in their community, including their schools, for example peers or colleagues, friends, or religious leaders; and between students/teachers and health professionals.

However, it is possible that questioning other people's claims or beliefs—especially those of authority figures—will cause conflict. We identified this potential effect in the process evaluation for the trial of the primary school resources, but it was not reported or observed in the process evaluation (Nsangi *et al.*, 2019) or the 1-year follow-up of the trial (Nsangi *et al.*, 2020). A minority of participating teachers have raised it as a potential adverse effect in the prioritisation of IHC Key Concepts for the secondary school resources.

Opportunity cost harms

Waste of time and resources

Using the IHC secondary school resources should be worthwhile in practical and academic terms, for students and teachers, and their schools as a whole, both in actuality and from the users' perspectives. However, if the resources are ineffective or if what the students learn is perceived as inessential, it is possible that the resources will in fact or perception be a waste of time and resources. This might include a negative effect on performance in standard subjects, and thereby negative effects on the opportunities of students, performance appraisals of teachers, and reputations of schools.

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